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GRAPHICAL DATA DISPLAY With Arduino And HTML5

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Most Arduino-based projects use LCDs to show data, but this project instead uses webpage so that more details can be displayed in a graphical form and accessed from anywhere through the Internet. The project takes variable analogue inputs from Arduino and displays results in graphical form on a webpage that supports HTML5.

The project presented here includes four webpages coded in four .htm files, which are stored in a memory card and then hyperlinked with each other.

Fig. 1: Ethernet shield with microSD card

A web server created by an Arduino Ethernet shield loads the requested webpage from an SD card memory device into the browser. The webpage issues AJAX requests to the web server at regular intervals to get input data. The web server responds with an XML file containing measured input values in appropriate tags. The webpage reads input values from the XML format using JavaScript and displays them in an appropriate diagram/chart coded with HTML5 canvas element.

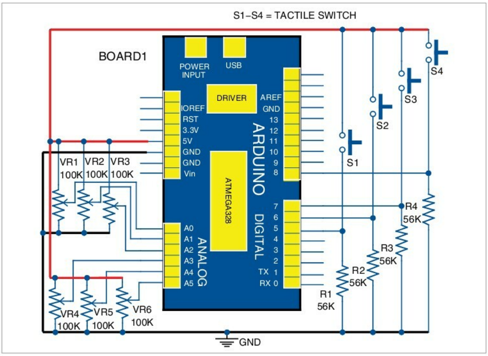

Fig. 2: Circuit diagram for graphical data display with Arduino and HTML5

A canvas is a drawable region defined in HTML5 code specified by <canvas> tag for dynamically generating graphics. Animation of the graphics is possible using JavaScript code. Canvas enables programmers to create visually appealing drawings with a few lines of code. It is especially suited for Arduino because of its limited capability to deal with large files.

Circuit and working

The Ethernet shield with microSD card is shown in Fig. 1. The circuit for Arduino-based graphical data display is shown in Fig. 2. It is built around an Arduino Uno board along with some potentiometers (VR1 through VR6), resistors (R1 through R4) and tactile switches (S1 through S4).

Arduino web server. The web server is implemented by stacking Ethernet shield on top of Arduino board. The Ethernet shield used in this project has W5100 Ethernet controller and an onboard microSD card slot. Transfer HTML5 files required for this project (gauge.htm, bargraph.htm, analog.htm and digital.htm) to an SD card. Now insert the SD card in the onboard microSD card slot of Ethernet shield.

Arduino communicates with both W5100 and SD card using the SPI bus through the ICSP header, and hence only one of these can be active at a time. Arduino sketch (Arduino_html5.ino) in this project uses SPI, Ethernet and SD libraries to function. These libraries manage sharing of the SPI bus so that both W5100 and SD card can be utilised.

Connecting Ethernet shield to the PC. Connect Ethernet shield to the PC using Ethernet cable. The IP address assigned to Ethernet shield in Arduino